

eSSL Device Commissioning

Getting a biometric machine from its box to pushing attendance into payroll — and diagnosing it when it doesn't.

Applies to eSSL / ZKTeco / Matrix units with ADMS push Protocol iclock ADMS over HTTP(S)

Verified 31 Aug 2026

00 Before you go to site

OFFICE · 10 MINUTES

Three things decide whether this install takes twenty minutes or a return visit. Settle them before you travel.

Confirm the device actually pushes

The machine must support **ADMS** — eSSL also labels it *Cloud Server*, *Webserver*, or *WebSetup*. On the spec sheet it appears as **Push Data: Yes**. If the only connectivity options are a USB stick, RS232/485, or "PC software / SDK", the unit cannot work with StatComPy.

Face, fingerprint, card, and password units are all equally fine. The machine only ever sends us a User ID and a timestamp — never the biometric template, which stays on the device.

Align employee codes — the step people skip

The device sends only the **User ID** you assign each person at enrolment. StatComPy matches that ID to `employee_code` as an **exact string**: trimmed of spaces, case-sensitive, no zero-padding, no prefix stripping. `EMP0042` and `42` are different people as far as the match is concerned.

"USER ID", NOT A PIN ANYONE TYPES

eSSL's data format labels this field *PIN*, which is misleading. Nobody enters it at the door — employees punch with face or finger only, and the device can be set to face-only or finger-only so the keypad option never appears. The User ID is filled in once, by whoever does enrolment, and the device stamps it onto every punch automatically once it recognises the person. It is what lets a face match become a name in payroll.

Check now whether the device accepts alphanumeric user IDs. Many eSSL units are numeric-only. If yours is, and your HR master uses `EMP0042`, every punch

will arrive unmatched — recoverable, but you don't want to find out after enrolling three hundred people.

Check the network path

The device initiates every upload. There is no polling from our side, so if the site firewall blocks outbound traffic to `app.statcosol.com`, nothing arrives and nobody is alerted. Confirm outbound 443 (or 80) is open from the VLAN the device will sit on.

☐ Spec sheet or vendor confirms

ADMSupport, in writing

/

Push

Data

☐ Device serial number photographed or noted (printed on the back)

☐ Employee codes in StatComPy match what the device can accept

☐ Outbound 443 open from the device VLAN to

`app.statcosol.com`

☐ Branch that this device belongs to is decided

TELL THE INSTALLER

Do **not** let the vendor install eTimeTrackLite or point the device at eBioServer.

An ADMS device pushes to one server address only — if it is aimed at eSSL's own software, it will not reach StatComPy.

01 Register the device in StatComPy

CLIENT PORTAL · DO THIS FIRST

ORDER MATTERS

Register the serial number **before** configuring the machine. An unregistered serial is rejected with a 401, and the device retries silently forever — no error on the device, no alert in StatComPy. It simply looks like nothing is happening.

Sign in to the client portal and open **Payroll & Workforce** → **Biometric Devices**. On the **Devices** tab, click **+ Add Device**.

Register Biometric Device

SERIAL NUMBER *

ABCD1234567

exactly as printed

LABEL

Main Gate K90

BRANCH

Chennai — Ambattur Unit

VENDOR

eSSL

MODEL

K90

Schematic, not a screenshot. Field names and order match the live form.

Serial Number is the only required field, and it cannot be edited afterwards — delete and re-add if you mistype it. Vendor offers eSSL, ZKTeco, Matrix, and Other; Model is free text.

Saving creates the device row as **Enabled** and generates a push token. You do not need the token: the `/iclock` endpoint authenticates on serial number alone today.

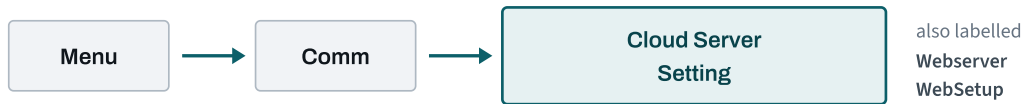
KNOWN GAP

Because authentication is serial-number-only, anyone who learns a registered serial can post punches for that client — and those punches auto-approve into payroll. **Rotate Token** does not yet revoke anything at the device boundary. Treat serial numbers as semi-sensitive and don't circulate them beyond the install team.

02 Configure the machine

AT THE DEVICE · ONE TIME

On the keypad: **Menu** → **Comm** → **Cloud Server Setting**. The label varies by model — on some units the same screen sits under **Comm** → **Webserver** or **WebSetup**.



Navigation path on the device keypad. If none of these three labels exists anywhere under Comm, the unit does not support ADMS push.

Enter these values

FIELD	VALUE	WHY
Server Address	app.statcosol.com	The push host
Server Path	/iclock	Often blank by default; the device will not connect without it
Server Port	443	Or 80 — see the fallback below
HTTPS / SSL	Enable	Disable only if the unit has no TLS support
Server Mode	ADMS	Some models label this "HTTP" or "Cloud"
Enable Domain Name	Yes	We publish a hostname, not a fixed IP
Realtime Upload	Yes	Push each punch as it happens
Heartbeat	10 sec	Default; leave it alone
Proxy	None	

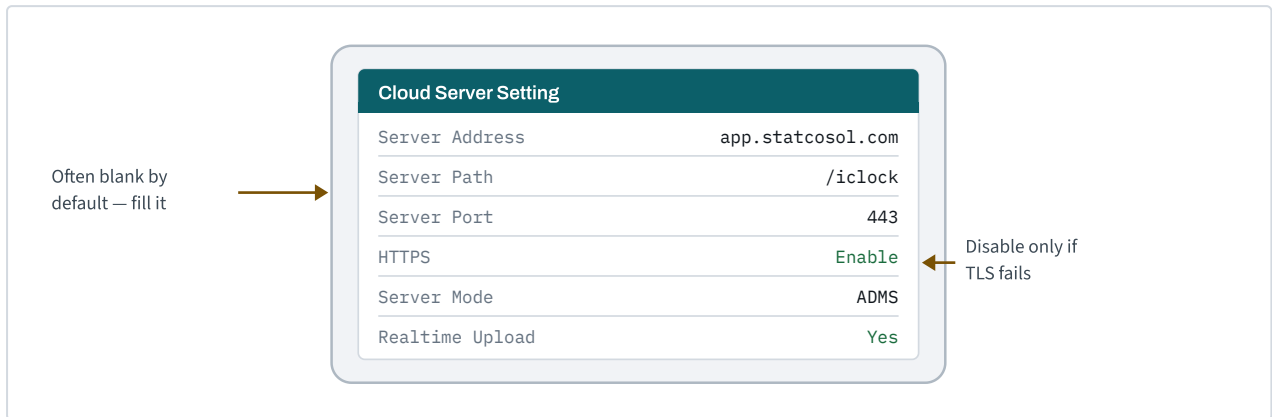
PORT 80 IS A REAL FALLBACK

Many older eSSL units have no usable TLS stack and fail *silently* against HTTPS — no error, just nothing arriving. Both schemes are served and neither redirects, measured 31 Aug 2026:

```
$ curl -s -o /dev/null -w "%{http_code}\n" http://app.statcosol.com/iclock/ping
200
$ curl -s -o /dev/null -w "%{http_code}\n" https://app.statcosol.com/iclock/ping
200
```

Prefer 443. Drop to port 80 with SSL disabled only when the handshake fails — punches carry an employee code and a timestamp, so plaintext is a real if limited exposure.

Save and **reboot the device**. Within about thirty seconds it calls `GET /iclock/cdata?SN=...&options=all`, and the backend replies with the session configuration — timezone IST, realtime on, heartbeat interval. From then on every punch is pushed within the heartbeat window.



Schematic of the completed screen. Layout and typography differ by model — match the values, not the appearance.

03 Enrol employees on the device

AT THE DEVICE · PER PERSON

On the device: **User Mgmt** → **New User**. Set the **User ID** to the person's StatComPy `employee_code`, then capture their face, finger, or both. Everything else — the name stored on the device, card assignment — is local to the machine and never reaches us.

THE ONE RULE

User ID on device = `employee_code` in StatComPy. Exact match, character for character.

Enrol **both face and finger** where the unit supports it. It costs nothing — capacity is 3,000 of each — and it saves you when face matching struggles: bright backlight at a gate, a mask or helmet, or simply a face the camera dislikes. That person uses a finger instead of standing there being rejected.

If a code doesn't match, the punch is not lost. It is stored with no employee linked, shows as **Unknown** in the punch feed, and can be recovered later: fix the code in HR master, then click **Reconcile Unknown** on the Devices tab. Past punches are re-linked and the attendance days recomputed.

Enrol two or three people first and verify one punch end to end before working through the whole roster. A mismatch found at person three costs minutes; found at person three hundred it costs an afternoon.

04

Verify before you leave site

5 MINUTES · DO NOT SKIP

Run this from the site network, not from the office — you are testing the path the device will actually use.

```
$ curl https://app.statcosol.com/iclock/ping
OK
```

Then punch a finger or face on the device and, within about ten seconds, open **Biometric Devices** → **Punch Feed** in the client portal with today's date range.

COLUMN	WHAT YOU SHOULD SEE
Time	The moment you punched, in local time
Employee Code	The User ID you enrolled — matching HR master exactly
Device (SN)	The serial number you registered
Direction	IN OUT or AUTO depending on the key pressed
Linked	✓ — a green tick. Unknown means the code doesn't match
Processed	✓ — rolled up into the attendance day

Back on the **Devices** tab, the row should now show a **Last Seen** timestamp and a non-zero **Punches** count. A device that has never connected reads Never.

☐ /iclock/ping

return OK

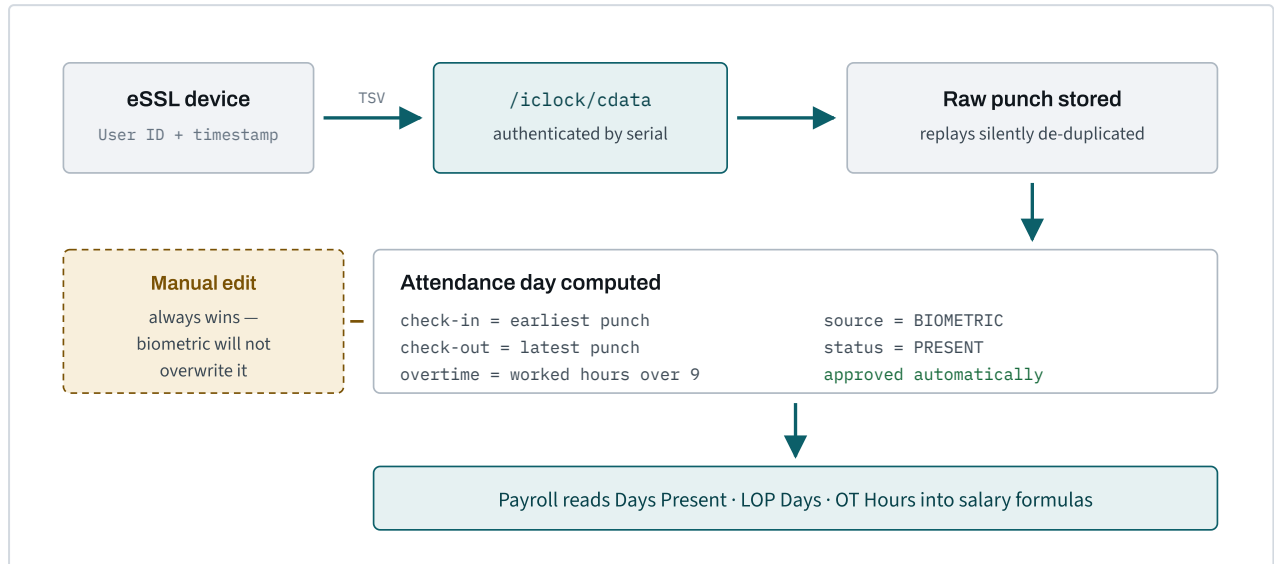
from
the
site
network

- ☐ Test punch appears in the Punch Feed within ~10 seconds
- ☐ Linked shows a green tick, not Unknown
- ☐ Devices tab shows Last Seen and a punch count
- ☐ The day appears in Daily Attendance with a check-in time

What happens to a punch

REFERENCE · NO ACTION NEEDED

Useful when someone asks why a number looks the way it does.



Overtime is fixed at hours worked beyond nine per day. A day edited by hand in Mark Attendance is locked against later device pushes, so a bad scan can always be corrected.

NO REVIEW STEP ON AN ESSL SITE

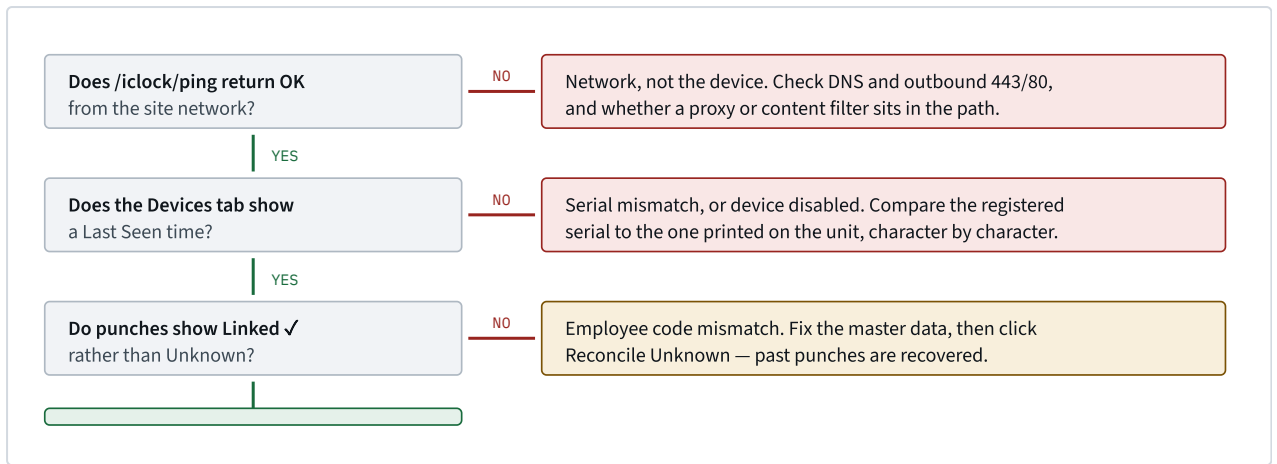
A client runs **either** eSSL devices **or** the FaceDesk kiosk, never both. On an eSSL site the day is approved on arrival and reaches payroll with no human step — the device did the matching and we trust it.

The exception is a **migration** between the two, when both are briefly live. On any day containing a FaceDesk face punch, the whole day goes to the review queue instead of auto-approving. Expect a short spike in the queue over a cutover, and make sure somebody clears it — unreviewed days do not reach payroll.

When nothing arrives

DIAGNOSIS IN ORDER

Work down this list. Each step rules out one layer.



Each "no" branch identifies the layer at fault. Reaching the bottom means punches are arriving and linked — any remaining problem is in attendance or payroll configuration, not the device.

SYMPTOM	CAUSE AND FIX
Device says "upload failed"	Check DNS and outbound reach. If it fails only on 443, the firmware likely lacks TLS or SNI — switch to port 80 with SSL off.
Nothing at all, no error anywhere	Serial not registered. The device retries silently against a 401 forever.
Punches arrive, employee blank	User ID ≠ <code>employee_code</code> . Fix the master data, then Reconcile Unknown .
Times off by 5h30m	Device timezone wrong. Set it to IST on the machine, or override <code>tzOffsetMinutes</code> on the device row.
Check-in but no check-out	Only one punch that day — the person forgot to punch out. Correct it in Mark Attendance.
OT hours showing zero	Worked hours are 9 or under, or the attendance day is not approved.
A manual edit got overwritten	The manual row was missing a check-in. Set both check-in and check-out to lock it.



Running it afterwards

HR · ROUTINE

WHEN	DO THIS
New employee joins	Enrol on the device using the same code as their <code>employee_code</code> .
Code was mistyped	Fix the master data, then Reconcile Unknown on the Devices tab.

WHEN	DO THIS
Site loses internet	Nothing. The device buffers in flash — 100,000+ punches — and pushes everything on reconnect. Replays are de-duplicated.
A scan was wrong	Edit the day in Mark Attendance. The row becomes MANUAL and is locked against later pushes.
Backfill a date range	Reprocess Range on the Punch Feed tab, after fixing whatever was wrong.
Machine retired	Disable the device row, or delete it. Disabling stops new pushes and keeps the history.

ADDING A SECOND MACHINE

Register each unit separately with its own serial. Several devices can serve one branch — punches merge into the same attendance day, with the earliest becoming check-in and the latest check-out, regardless of which machine recorded them.

Endpoint behaviour and the port-80 fallback were verified against the live service on 31 August 2026. Device menu labels vary between eSSL models — the values in this procedure are what matter, not the wording on any particular screen.

Diagrams in this document are schematics drawn to match the live forms and screens; they are not photographs of a specific unit.